

Learning

Let D be a set of ^{iid} samples from a probability distribution. The probability $P(D: \mathcal{M}) = \prod P(x^{[m]}: \mathcal{M})$

is the **training set likelihood**. We can evaluate a model using the **test set likelihood**. Both a function $L(\theta: D)$ of the model parameters.

A function $s: \text{Val}(X) \rightarrow \mathbb{R}^k$ is a sufficient statistic if for all samples D, D' , we have $s(D) := \sum_{x \in D} s(x) = s(D') \Rightarrow L(\theta: D) = L(\theta: D')$.

In a multinomial distribution with k values, the counts $M_i = \#x^i$ are sufficient statistics. We have $s: x^i \mapsto (0, \dots, 0, 1, 0, \dots, 0)$, so $\sum s(x^{[m]}) = (M_1, \dots, M_k)$.

In a Gaussian distribution $P \sim \mathcal{N}(\mu, \sigma^2)$, $s(x) = \{1, x, x^2\}$ is a sufficient statistic.

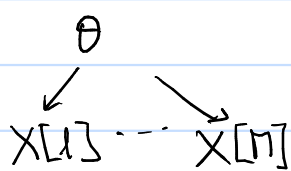
The **maximum likelihood estimation (MLE)** is $\hat{\theta} = \text{argmax}_{\theta} L(\theta: D)$.

For Bayesian networks,

$$\begin{aligned} L(\theta: D) &= \prod_m P(x^{[m]}: \theta) \\ &= \prod_m \prod_i P(x_i^{[m]} | \text{Par}_i x_i^{[m]}: \theta_i) \\ &= \prod_i L_i(\theta: D). \end{aligned}$$

If $\theta_{x_i}, \text{Par}_i x_i$ are disjoint then we can maximise these local likelihoods individually.

In Bayesian estimation, we assume that θ is a random variable. We get a probabilistic model



with $P(x[1], \dots, x[M], \theta) = P(\theta) P(x[1], \dots, x[M] | \theta)$

$$= P(\theta) \prod P(x[i] | \theta) = P(\theta) L(\theta: D).$$

We have $P(\theta | D) = \frac{L(\theta: D) P(\theta)}{P(D)}$

$\swarrow$ **posterior** $\nwarrow$ **prior**
 $\nwarrow$ **constant rel. to θ**

We say that θ follows a **Dirichlet distribution**, $\theta \sim \text{Dirichlet}(\alpha_1, \dots, \alpha_k)$, if

$$P(\theta) = \frac{1}{Z} \prod_{i=1}^k \theta_i^{\alpha_i - 1}$$

where $Z = \frac{\prod \Gamma(\alpha_i)}{\Gamma(\sum \alpha_i)}$ and $\theta \in [0, 1]$.

We have $P(\theta | D) \propto L(\theta: D) P(\theta)$

$\nwarrow$ $\prod \theta_i^{M_i}$ $\nwarrow$ $\propto \prod \theta_i^{\alpha_i - 1}$
 $\nwarrow$ $\sim \text{Dir}(\alpha_1 + M_1, \dots, \alpha_k + M_k)$

prior & posterior come from the same family of distributions

We say that Dirichlet is a **conjugate prior** for the multinomial distribution.

We have $P(X = x^i) = \int_{\theta} P(x^i | \theta) P(\theta) d\theta = \frac{1}{Z} \int_{\theta} \theta_i \cdot \prod \theta_j^{\alpha_j - 1} d\theta = \frac{\alpha_i}{\sum \alpha_j}$.

Now, $P(x[M+1] | x[1], \dots, x[M])$

$$\begin{aligned} &= \int_{\theta} P(x[M+1] | x[1:M], \theta) P(\theta | x[1:M]) d\theta \\ &= \int_{\theta} P(x[M+1] | \theta) P(\theta | x[1:M]) d\theta. \end{aligned}$$

$\nwarrow$ **posterior $\sim \text{Dir}(\alpha_i + M_i)$**
 $\nwarrow$ θ_i $\nwarrow$

Hence, $P(x[M+1] = x^i | x[1:M]) = \int_{\theta} P(x[M+1] = x^i | \theta) P(\theta | x[1:M]) d\theta$

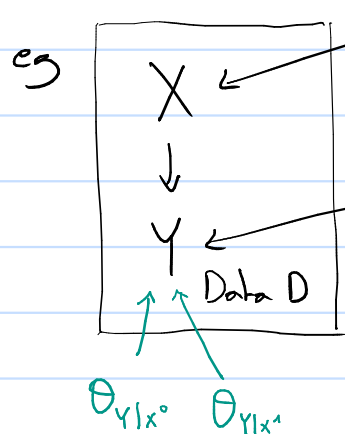
$$= \frac{\alpha_i + M_i}{\sum (\alpha_j + M_j)}$$

similarly as above. We call $\alpha = \sum \alpha_i$ the **equivalent sample size**. Compare this to the MLE, where the probability would be $\frac{M_i}{\sum M_j}$.

Let $\theta \sim \text{Unif}(0, 1) = \text{Dirichlet}(1, 1)$. Let $M_0 = 1, M_4 = 4$. The Bayesian estimation of $P(X[6] = 1)$ is $5/7$, which is smaller than the MLE of $4/5$.

Asymptotically, both estimations agree.

Bayesian estimation in Bayesian networks



- $(X^{[m]}, Y^{[m]} \perp X^{[m]}, Y^{[m]} | \theta)$
- $P(\theta) = \prod P(\theta_{x_i} | \text{Par}_i x_i)$
- Posteriors of θ are independent given D
 $P(\theta_{x_i}, \theta_{Y|X} | D) = P(\theta_{x_i} | D) P(\theta_{Y|X} | D)$
- $(\theta_{Y|X^0} \perp \theta_{Y|X^1} | X[1], Y[1])$ even though $Y[1]$ activates a v-structure

In general, assume X is multinomial or $\theta \sim \text{Dir}(\alpha_{x|u}, \dots, \alpha_{x^*|u})$. Then the posterior is $\text{Dir}(\alpha_{x|u} + M_i[\text{Par}_X], \dots, \alpha_{x^*|u} + M_k[\text{Par}_X])$. To specify $\alpha_{x|u}$, we start with a prior network with parameters θ_0 (e.g. uniformly chosen) and an equivalent sample size parameter α , and set $\alpha_{x|u} = \alpha P(x, u | \theta_0)$.

$X \downarrow Y$

$P(X = x^i | \theta_0) = \frac{1}{2} \Rightarrow \alpha_x = \frac{\alpha}{2} \Rightarrow \theta_x \sim \text{Dir}(\frac{\alpha}{2}, \frac{\alpha}{2})$

$P(Y = y^i, X = y^j | \theta_0) = \frac{1}{4} \Rightarrow \alpha_{Y|X} = \frac{\alpha}{4} \Rightarrow \theta_{Y|X} \sim \text{Dir}(\frac{\alpha}{4}, \frac{\alpha}{4})$

Log-Likelihood for Markov networks

$$P_{\theta}(a,b,c) = \frac{1}{Z} \phi_1(a,b) \phi_2(b,c)$$

A
↓
B
↓
C

$$\Rightarrow \ell(\theta; D) = \sum \log \phi_1(a^{[m]}, b^{[m]}) + \log \phi_2(b^{[m]}, c^{[m]}) - \log Z(\theta)$$

$$= \sum_{a,b} M[a,b] \log \phi_1(a,b) + \sum_{b,c} M[b,c] \log \phi_2(b,c) - M \log Z(\theta)$$

$$\text{with } Z(\theta) = \sum_{a,b,c} \phi_1(a,b) \phi_2(b,c)$$

$\Rightarrow$ the partition function couples the parameters, we cannot maximise ϕ_1, ϕ_2 individually!

Let's consider a log-linear model $P(X_1, \dots, X_n; \theta) = \frac{1}{Z(\theta)} \exp \sum \theta_i f_i(D)$.

Then $\ell(\theta; D) = \sum \theta_i \left(\sum f_i(x^{[m]}) \right) - M \log Z(\theta)$,

where $\log Z(\theta) = \log \sum \exp \left(\sum \theta_i f_i(x) \right)$.

Prop: $\frac{\partial}{\partial \theta_i} \log Z(\theta) = E_{\theta}[f_i]$, $\frac{\partial^2}{\partial \theta_i \partial \theta_j} \log Z(\theta) = \text{Cov}_{\theta}[f_i, f_j]$.

In particular, $\log Z(\theta)$ is convex. Hence, $\ell(\theta; D)$ is concave as the sum of a concave and a linear function.

We have $\frac{\partial}{\partial \theta_i} \frac{1}{M} \ell(\theta; D) = \overset{\text{empirical expectation}}{E_D[f_i(X)]} - \overset{\text{expectation wrt } P_{\theta}}{E_{\theta}[f_i]}$.

Hence, $\hat{\theta}$ is the MLE if and only if

$$E_D[f_i(X)] = E_{\hat{\theta}}[f_i] \quad \forall i.$$

To find the MLE we typically use L-BFGS, a variant of gradient ascent using a second order instead of a first order approximation. To get a gradient, we need the expected feature counts $E_D[f_i(X)]$ and the inferences $E_{\theta}[f_i]$.

Example (Ising model): $E(x_1, \dots, x_n) = - \sum_{i,j} w_{ij} x_i x_j - \sum_i u_i x_i$ $\uparrow$ ± 1

$$\frac{\partial}{\partial u_i} \frac{1}{M} \ell(\theta; D) = \frac{1}{M} \sum x_i^{[m]} - (P_{\theta}(X_i=1) - P_{\theta}(X_i=-1))$$

$$\frac{\partial}{\partial w_{ij}} \frac{1}{M} \ell(\theta; D) = \frac{1}{M} \sum x_i^{[m]} x_j^{[m]} - (P_{\theta}(X_i=1, X_j=1) + P_{\theta}(X_i=-1, X_j=-1) + P_{\theta}(X_i=1, X_j=-1) + P_{\theta}(X_i=-1, X_j=1))$$

$\nwarrow$ conditional random field
For CRFs, we have $P_{\theta}(Y|x) = \frac{1}{Z_x(\theta)} \tilde{P}_{\theta}(x, Y)$, $Z_x(\theta) = \sum_Y \tilde{P}_{\theta}(x, Y)$.

We want to maximise

$$\ell_{Y|x}(\theta; D) = \sum \log P_{\theta}(y^{[m]} | x^{[m]}, \theta).$$

In the case of a log-linear model, we get

$$\frac{\partial}{\partial \theta_i} \frac{1}{M} \ell_{Y|x}(\theta; D) = \frac{1}{M} \sum (f_i(x^{[m]}, y^{[m]}) - E_{\theta}[f_i(x^{[m]}, Y)])$$

Note that we have to run an inference for each sample $x^{[m]}$, but we only need $P(Y|x)$ instead of $P(Y, X)$.

MAP learning

Gaussian prior $P(\theta; \sigma^2) = \prod \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{\theta_i^2}{2\sigma^2}\right)$ $\nwarrow$ hyperparameter

Laplacian prior $P(\theta; \beta) = \prod \frac{1}{2\beta} \exp\left(-\frac{|\theta_i|}{\beta}\right)$

We want to find

$$\arg\max_{\theta} P(D, \theta) = \arg\max_{\theta} P(D|\theta) P(\theta)$$

$$= \arg\max_{\theta} \ell(\theta; D) + \log P(\theta).$$

quadratic for Gaussian
l.l for Laplacian
 $\Rightarrow L^2, L^1$ -regularisation, resp.

Structure Learning

Note that if a Bayesian network is missing an edge then P^* cannot be learned. Spurious dependencies, created when adding edges, lead to an increased parameter space and can worsen generalisation.

We define the **likelihood score** of G given D as

$$\text{score}_L(G: D) = \ell(\hat{\theta}: D) \quad \text{MLE} \quad \text{argmax}_{\theta} \ell(\theta: D)$$

$$\begin{aligned} G_1: X \rightarrow Y & \quad G_2: X \rightarrow Y \\ \text{score}_L(G_1: D) &= \sum_{x,y} \log \hat{\theta}_{x|y} + \log \hat{\theta}_{y|y} \quad \text{score}_L(G_2: D) = \sum_{x,y} \log \hat{\theta}_{x|y} + \log \hat{\theta}_{y|y} \\ \text{score}_L(G_2: D) - \text{score}_L(G_1: D) &= \sum_{x,y} \log \hat{\theta}_{y|y} - \log \hat{\theta}_{y|y} \\ &= \sum_{x,y} M[x,y] \log \hat{\theta}_{y|x} - \sum_y M[y] \log \hat{\theta}_y \quad \text{empirical distribution} \\ &= M \left(\sum_{x,y} \hat{P}(x,y) \log \hat{P}(y|x) - \sum_y \hat{P}(y) \log \hat{P}(y) \right) \\ &= M \sum_{x,y} \hat{P}(x,y) (\log \hat{P}(y|x) - \log \hat{P}(y)) \\ &= M \sum_{x,y} \hat{P}(x,y) \log \frac{\hat{P}(y|x)}{\hat{P}(y)} = M I_P(X,Y) \end{aligned}$$

We call $I_P(X,Y) = \sum_{x,y} P(x,y) \log \frac{P(y|x)}{P(y)} = \sum_{x,y} P(x,y) \log \frac{P(x,y)}{P(x)P(y)}$

the **mutual information**. The likelihood score decomposes as

$$\text{score}_L(G: D) = M \sum_i I_{\hat{P}}(X_i, \text{Par } X_i) - H_{\hat{P}}(X_i),$$

where $H_P(X) = -\sum P(x) \log P(x)$ is the **entropy** and is independent of G .

In particular, $\text{score}_L(G: D)$ is higher when X_i and $\text{Par } X_i$ are correlated.

Recall that $\text{score}_L(G_2: D) - \text{score}_L(G_1: D) = M I_P(X,Y),$

which is zero iff $X \perp Y$. This will probably not be true in $\hat{P}$, so $I_{\hat{P}}(X,Y) > 0$ almost always. In general, $\text{score}_L(G: D)$ is maximised in the fully connected graph.

To avoid overfitting, we

- Restrict the hypothesis space
$\text{Par } X_i$, or # θ
- Use a score that penalises complexity directly, or a Bayesian score

We define the **BIC score** as $\text{score}_{BIC}(G: D) = \ell(\hat{\theta}: D) - \frac{\log M}{2} \dim G.$

grows linearly with M (for $\ell(\hat{\theta}: D)$)
indep. parameters (for $\dim G$)
grows logarithmically with M (for $\frac{\log M}{2}$)

Its negative is called the **minimum description length (MDL)**.

A score sc is **consistent** if $sc(G^*: D) \rightarrow \max_G sc(G: D)$ for G^* the true structure or any I-equivalent structure. Note that both score_L and score_{BIC} are consistent.

We define the **Bayesian score** as $\text{score}_B(G: D) = \log P(D|G) + \log P(G).$

We have $P(D|G) = \int P(D|G, \theta) P(\theta|G) d\theta = \int \ell(\theta: D) P(\theta|G) d\theta.$

We also have $P(D|G) = \prod P(x[i] | x[< i], G).$

This has an explicit description and can be easily calculated on a computer.

For $P(G)$, we need to fix a prior over G :

- uniform $P(G) \propto \text{const.}$
- penalise edges $P(G) \propto c^{|G|}$ for $0 < c < 1$
- penalise #parameters

For $P(\theta|G)$, we can use the **BDe prior** by picking an equivalent sample size α and a BN B_0 representing the prior probability of events, and set

$$\alpha(x_i, \text{Par}_G x_i) = \alpha P(x_i, \text{Par}_G x_i | B_0).$$

We can use the BDe prior for all candidates G of G^* . It is the unique prior such that I-equivalent graphs have the same scores.

As $M \rightarrow \infty$, a network G with Dirichlet priors satisfies

$$\begin{aligned} \log P(D|G) &= \ell(\hat{\theta}: D) - \frac{\log M}{2} \dim G + O(1) \\ &= \text{score}_{BIC}(G: D) + O(1). \end{aligned}$$

Forests

A BN is a **forest** if $\# \text{Par } X \leq 1 \forall X \in G$. In this case, let $p(i)$ denote the parent of X_i (or 0). Then

$$\begin{aligned} \text{score}(G: D) &= \sum_{p(i) \neq 0} \text{score}(X_i | X_{p(i)}: D) + \sum_{p(i) = 0} \text{score}(X_i: D) \\ &= \sum_{p(i) \neq 0} (\text{score}(X_i | X_{p(i)}: D) - \text{score}(X_i: D)) + \sum_i \text{score}(X_i: D) \end{aligned}$$

score of an edge $p(i) \rightarrow X_i$
same for all trees (score of the empty network)

Let $w(i \rightarrow j) = \text{score}(X_j | X_i) - \text{score}(X_j).$

For the likelihood score, $w(i \rightarrow j) = M I_P(X_i, X_j) \geq 0$. Hence, the optimal structure is always a tree. For BIC and BDe, weights can be negative and the optimal structure may be a proper forest.

A score satisfies **score equivalence** if I-equivalent structures have the same score. In particular, such a score has $w(i \rightarrow j) = w(j \rightarrow i)$. We can therefore use undirected graphs. We get the following algorithm:

- Define an undirected graph with nodes $\{1, \dots, n\}$
- Set $w(i,j) = \max(\text{score}(X_j | X_i) - \text{score}(X_j), 0)$
- Remove weight-zero edges
- Find a forest of maximal weight (in $O(n^2)$)

General graphs

Theorem: Finding a maximally scoring network structure with at most k parents for each variable is NP-hard for $k \geq 1$.

Algorithm (Greedy hill climbing):

- Fix a starting network
- Add/delete/reverse any edge that improves the score the most

Problems:

- local maxima
- plateaux (typically because of equivalent networks)

Possible fixes:

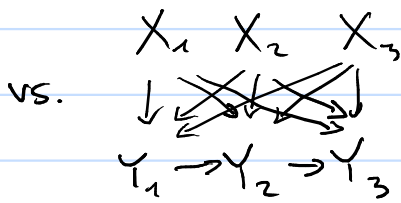
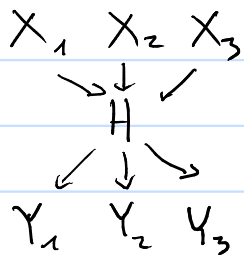
- Take a number of random steps
- Keep a taboo list (don't reverse the last k steps)

Advantages:

- The score decomposes as a sum of $\text{score}(X | \text{Par } X)$. Since we are only interested in Δscore , most terms will cancel at a step.

Incomplete data (Hidden (unobserved) variables, missing values)

Latent (unobserved) variables can lead to sparser models:

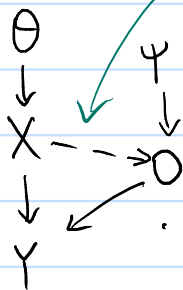


They may also point to clusters in the data.

0 or 1 depending on if X_i was observed

Let $X = \{X_1, \dots, X_n\}$ be a set of random variables, $\theta = \{\theta_1, \dots, \theta_n\}$ the binary observability variables, $Y = \{Y_1, \dots, Y_n\}$ the random variables with $\text{Val}(Y_i) = \text{Val}(X_i) \cup \{?\}$ and $Y_i = \begin{cases} X_i & \text{if } \theta_i = 1 \\ ? & \text{otherwise} \end{cases}$.

We have



deliberate missing data

We have the **missing at random** (MAR) assumption if

$$P_{\text{missing}} \neq (O \perp\!\!\!\perp H | \mathcal{Z})$$

unobserved X_i

observed Y_i

We are going to assume MAR.

With hidden variables or missing data, we often get multiple global maxima of the likelihood (e.g. by relabelling hidden variables).

expensive computation

Theorem:
$$\frac{\partial}{\partial \theta_{x_i|u_i}} P(D|\theta) = \frac{1}{\theta_{x_i|u_i}} \sum_u P(x_i, u | \mathcal{Z}[u], \theta)$$

Algorithm (Expectation maximisation):

1. Initialise θ
2. Infer missing data from θ and D to get D'
3. Estimate parameters (MLE) using D'

inferring data from θ is inexpensive

maximisation using complete data is inexpensive

Details:

- Instead of sampling D' , only compute $P(X, U | \mathcal{Z}[u], \theta)$
- Compute the **expected sufficient statistics** for each x, u

$$\bar{M}_\theta[x, u] = \sum_u P(x, u | \mathcal{Z}[u], \theta)$$

- Treat $\bar{M}_\theta$ as real and compute $\theta'_{x|u} = \frac{\bar{M}_\theta[x, u]}{\bar{M}_\theta[u]}$

Intuitively, at each step, the EM-algorithm optimises an expected MLE. Hence,

$$\ell(D: \theta^{t+1}) \geq \ell(D: \theta^t),$$

and if $\theta^{t+1} = \theta^t$ then $\ell(D: \theta)$ is a stationary point.

Final exam: incorrect at

6, 9, 10